

Detecting and Predicting Diabetic Retinopathy

Literature Review

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1. Literature Review

1.1 Machine Learning and Classical Approaches

Diabetic retinopathy (DR) detection has been extensively studied using classical machine learning approaches applied to both retinal images and structured clinical data. Bhattacharjee et al. [1] classified DR from retinal fundus images using explicit feature engineering—extracting exudates, blood vessels, and microaneurysms—followed by SVM, Random Forest, and Naïve Bayes classifiers on a 13,402-image Kaggle dataset. Preprocessing involved green channel extraction and CLAHE contrast enhancement on 350×350 resized images. The Random Forest classifier achieved the best performance with 76.5% accuracy, 77.2% sensitivity, and 93.3% specificity across five DR stages; however, no explainable AI (XAI) technique was applied, limiting clinical interpretability. Similarly, Kim et al. [2] approached DR as a clinical prediction problem using structured survey data from KNHANES (N = 3,026), comparing logistic regression, XGBoost, decision tree, and random forest models. The random forest yielded the highest AUC of 0.748. Crucially, SHAP analysis was applied to identify key glycemic and disease-history features—such as HbA1c, fasting glucose, diabetes duration, and insulin use—as dominant predictors, offering a transparent, image-free risk stratification pipeline. Complementing this, Tsao et al. [3] evaluated classical ML algorithms on structured biomedical datasets containing clinical variables such as glucose levels, blood pressure, and demographics. Random Forest achieved accuracy exceeding 80%, and feature importance ranking was used to identify interpretable biomedical predictors, offering improved explainability compared to deep learning methods, though at the cost of capturing complex feature interactions.

1.2 Convolutional Neural Networks for DR Detection and Grading

Deep convolutional neural networks have become the dominant paradigm for automated DR grading from fundus images. Kalyani et al. [4] proposed a reformed Capsule Network for DR grading on the Messidor dataset (1,200 RGB images), addressing CNN pooling-related feature loss through dynamic routing. Images were resized to 512×512, with green channel extraction, CLAHE, and Gaussian filtering applied as preprocessing. Reported per-grade accuracies ranged from 97.6%–98.6% on a 75/25 train-test split, though optimization details (optimizer, learning rate, epochs) were not fully specified, limiting reproducibility. No XAI was applied. Tymchenko et al. [5] proposed a multi-stage transfer learning ensemble combining EfficientNet-B4, EfficientNet-B5, and SE-ResNeXt50 with extensive data augmentation on the APTOS 2019 dataset, achieving a quadratic weighted kappa score of 0.92546 with 0.99 sensitivity and specificity. SHAP was applied for post-hoc interpretability. Rao et al. [6] compared ResNet101 and DenseNet121 on 13,970 APTOS images (32×32 input), finding DenseNet121 superior at 98.99% accuracy versus ResNet101’s 92.37%, though the dataset combined both eyes without eye-specific labeling and no XAI was applied.

1.3 Graph Neural Networks and Topological Approaches

Beyond pixel-based CNNs, Sundar and Sumathy [7] introduced a hybrid graph convolutional framework for DR severity classification that explicitly models topological and relational structure in retinal images. The pipeline involved ROI segmentation using a VGG-16-style FCN, feature extraction via ImageNet-pretrained ResNet-18, variational autoencoder (VAE)-based feature learning, and final classification using a graph convolutional network (GCN). Experiments on Kaggle (3,464 images) and EyePACS (35,000 images) datasets—with rotation, zoom, and flip augmentation to address class imbalance—reported 90.34% accuracy, 97.54% sensitivity, and 89.56% specificity on EyePACS. Despite its graph-based relational framing, no dedicated XAI mechanism was applied.

1.4 Explainable AI for Clinical Transparency

Recognizing the clinical imperative for interpretability, Shahzad et al. [8] developed a transparent DR diagnosis framework integrating a deep CNN with LIME visual explanations for post-hoc interpretability. Trained on a 60/20/20 split (1,702 training samples) over 50 epochs, the model achieved 94% accuracy with a 6% miss rate for binary DR classification. LIME-highlighted

regions provided visual justification for model decisions, improving clinical trust and suitability for real-world healthcare deployment.

1.5 Foundation Models and Self-Supervised Learning

Large-scale pretraining has recently emerged as a transformative direction. Zhou et al. [9] proposed RETFound, a Vision Transformer (ViT-Large) pretrained via masked autoencoder (MAE) on 1.6 million unlabeled retinal images (CFP and OCT), achieving an AUROC of 0.943 on APTOS and 0.884 on MESSIDOR-2—representing a 15%+ improvement over commercial models. Explainability is supported through attention-map visualization; however, performance degrades on cohorts with distinct demographic profiles or imaging hardware. Arcadu et al. [10] applied CNN-based longitudinal learning from baseline fundus images to predict DR progression at future visits, reporting an AUC of approximately 0.79. The model captured subtle microvascular changes without manual feature engineering, though it operated as a black box without XAI. Bora et al. [11] scaled this approach using EyePACS (35,000+ images) to predict future DR risk, achieving an AUC of approximately 0.90—demonstrating strong generalization—but similarly omitted XAI mechanisms, limiting transparency.

1.6 Longitudinal and Multimodal Prediction Frameworks

The most recent advances address DR progression prediction using temporal and multimodal data. Zeghlache et al. [12] proposed L-MAE, a longitudinal masked autoencoder incorporating time-aware positional embeddings and disease-progression-aware masking on the OPHDIAT database (101,383 patients, 763,848 images, avg. 2.99 scans/patient over 2.23 years). L-MAE achieved an AUC of 0.798 for predicting severe DR at the next visit, significantly outperforming baseline longitudinal models (0.641 AUC), with progression-aware masking serving as an intrinsic XAI mechanism by identifying anatomically localized changes. Dai et al. [13] introduced DeepDR Plus, combining a deep CNN with Weibull mixture survival analysis for individualized five-year time-to-progression prediction. Pretrained on 717,308 images and validated on 118,868 images from 29,868 multiethnic participants, the model achieved a C-index of 0.754–0.846 and demonstrated clinical utility by potentially extending low-risk screening intervals from 12 to approximately 32 months; hazard-map visualizations served as the XAI component. Zhang et al. [14] introduced a graph-enhanced multimodal fusion framework combining a ResNet50 fundus branch and EfficientNet-B0 OCT branch, with structural-aware graph embeddings, SHAP, and Grad-CAM for explainability. Evaluated on a curated paired fundus-OCT dataset with CLAHE preprocessing (512×512), the model achieved 90.5% accuracy and an AUC-ROC of 0.970, with OCT integration overcoming 2D photography limitations in distinguishing mild and moderate NPDR stages.

2. Research Gap

Despite substantial progress, several critical gaps remain across the reviewed literature. First, the majority of single-modality models—whether image-based CNNs or clinical tabular classifiers—fail to leverage complementary information available in real clinical settings, where both retinal photographs and patient metadata coexist. Second, while XAI techniques (SHAP, LIME, Grad-CAM, attention maps) have been incorporated in select works, many high-performing models, including several deep learning approaches reviewed, remain opaque black boxes that clinicians cannot interpret or trust in deployment. Third, most studies frame DR detection as a static cross-sectional classification task, ignoring the inherently progressive and longitudinal nature of the disease; few works model temporal trajectories across multiple patient visits. Fourth, class imbalance across DR severity grades is frequently acknowledged but inconsistently addressed, with some works relying solely on augmentation while others omit this entirely, biasing reported metrics. Fifth, generalizability across diverse populations and imaging devices is rarely validated, with most models trained and tested on single-center or single-device datasets (e.g., APTOS, EyePACS, Messidor), limiting real-world applicability. Collectively, these gaps motivate the development of a multimodal, interpretable, and longitudinally-aware DR detection and prediction system that combines retinal image features with structured clinical data under a unified, explainable framework—as proposed in this work.

3. Comparative Analysis of Reviewed Methods

Reference	Algorithm	Dataset	Task	Best Metric	XAI	Modality	Limitation
Bhattacharjee et al. [1]	SVM, RF, Naïve Bayes	Kaggle (13,402 imgs)	5-class grading	RF: 76.5% Acc	No	Fundus image	No XAI; feature-engineered only
Kim et al. [2]	LR, RF, XGBoost, DT	KNHANES (3,026 pts)	Binary DR risk	RF: AUC 0.748	SHAP	Clinical tabular	No image data; severity not staged
Tsao et al. [3]	RF, SVM, LR, DT	Clinical biomedical	Binary DR risk	RF: >80% Acc	Feature rank	Clinical tabular	Limited complex feature interactions
Kalyani et al. [4]	Capsule Network	Messidor (1,200)	4-class grading	97.6–98.6% Acc	No	Fundus image	Incomplete training details
Tymchenko et al. [5]	EfficientNet ensemble	APTOS 2019	5-class grading	$\kappa = 0.925$	SHAP	Fundus image	High data cost; expert annotation
Rao et al. [6]	ResNet101, DenseNet121	APTOS (13,970)	5-class grading	DenseNet: 98.99%	No	Fundus image	No XAI; no eye-specific labeling
Sundar & Sumathy [7]	Hybrid GCN + VAE	Kaggle + EyePACS	5-class grading	90.34% Acc	No	Fundus image	No XAI despite graph framing
Shahzad et al. [8]	CNN + LIME	Kaggle (binary)	Binary detect.	94% Acc	LIME	Fundus image	Binary only; data source indirect
Zhou et al. [9]	ViT-L (RETFound)	1.6M unlabeled	Multi-task	AUROC 0.943	Attn maps	Fundus + OCT	Demographic generalizability drop
Arcadu et al. [10]	CNN (longitudinal)	Proprietary longit.	Progression pred.	AUC $\approx$ 0.79	No	Fundus image	Black box; no XAI
Bora et al. [11]	Deep CNN	EyePACS (35,000+)	Risk prediction	AUC $\approx$ 0.90	No	Fundus image	No XAI; transparency limited
Zeghlache et al. [12]	L-MAE (ViT)	OPHDIAT (763,848)	Progression pred.	AUC 0.798	Masking	Longitudinal fundus	Computational complexity
Dai et al. [13]	DeepDR + Weibull	717,308 (pretrain)	Time-to-progress	C-index 0.754–0.846	Hazard maps	Fundus image	Single-image; trajectory variance
Zhang et al. [14]	ResNet50 + EfficientNet-B0 + GCN	Paired fundus-OCT	5-class grading	AUC-ROC 0.970	SHAP + Grad-CAM	Fundus + OCT	Limited dataset diversity

Note: Acc = Accuracy; RF = Random Forest; LR = Logistic Regression; DT = Decision Tree; SVM = Support Vector Machine; κ = Quadratic Weighted Kappa; XAI = Explainable Artificial Intelligence; GCN = Graph Convolutional Network; VAE = Variational Autoencoder; ViT = Vision Transformer; NPDR = Non-Proliferative Diabetic Retinopathy.

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